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Date: 15.05.2020

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Int. to Financial Mathematics

20.225-

1) $A(t) = at^2 + bt + c$

$$A(0) = 200 \Rightarrow c = 200$$

$$A(2) = 280 \Rightarrow 4a + 2b + 200 = 280 \Rightarrow 2a + b = 40$$

$$A(3) = 390 \Rightarrow 9a + 3b + 200 = 390 \Rightarrow 3a + b = 50$$

$$a = 10$$

$$b = 20$$

$$A(t) = 10t^2 + 20t + 200$$

$$a) i_0 = \frac{A(3) - A(2)}{A(2)} = \frac{390 - 280}{280} = \frac{110}{280} = \frac{11}{28} = 0,392857$$

$$b) d = \frac{A(1,5) - A(0,5)}{A(1,5)} = \frac{(22,5 + 30 + 200) - (2,5 + 10 + 200)}{(22,5 + 30 + 200)} = \frac{40}{252,5} = 0,15841584$$

$$c) f = \frac{A'(t)}{A(t)} = \frac{20t + 20}{10t^2 + 20t + 200} = \frac{2t + 2}{t^2 + 2t + 20}$$

$$f_{1,5} = \frac{2 \cdot (1,5) + 2}{(1,5)^2 + 2 \cdot (1,5) + 20} = \frac{5}{25,25} = 0,198019802$$

d)


$$A(0) = 200 \quad A(1,5) = 252,5$$

$$\frac{A(0)}{A(1,5)} = \frac{x}{100}$$

$$\frac{200}{252,5} = \frac{x}{100}$$

$$x = \frac{20000}{252,5} = 79,20792079$$

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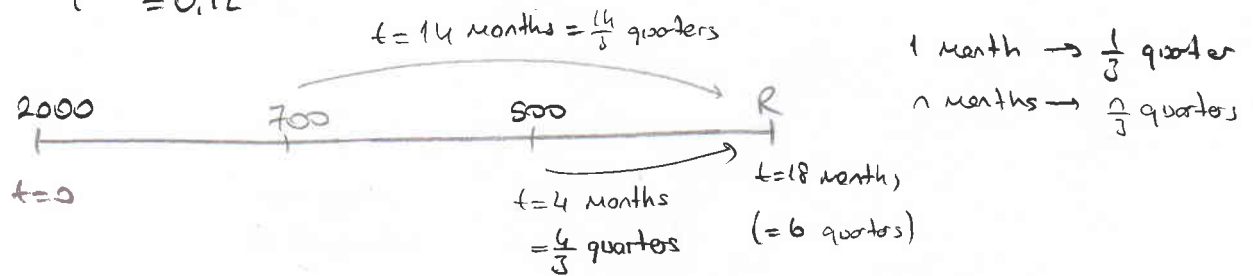
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2) convertible quarterly $\Rightarrow \left(1 + \frac{i^{(4)}}{4}\right)^t = \left(1 + \frac{0.12}{4}\right)^t = (1.03)^t$
 $i^{(4)} = 0.12$



$$700 \cdot (1.03)^{\frac{14}{3}} = 803.5355372$$

$$500 \cdot (1.03)^{\frac{4}{3}} = 520.0993419$$

$$2000 \cdot (1.03)^6 = 2388.104593$$

$$2388.104 = 803.5355372 + 520.0993419 + R$$

$$R = 1064.469714$$

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3) $1+i = 1 + (0,09) \cdot 2 = 1,18 \rightarrow$ (simple interest for two years)

$$\left(1 - \frac{(d)^{(2)}}{2}\right)^3 = \left(1 - \frac{0,08}{2}\right)^3 \rightarrow \left(\text{nominal discount convertible semiannually for 18 months (= 3 semiannual)}\right)$$
$$= (0,96)^3$$

$$(1 - dt)^{-1} = (1 - (0,1) \cdot (2,25))^{-1}$$
$$= (0,775)^{-1} \rightarrow \left(\text{simple discount for } \underbrace{2 \text{ year } 3 \text{ months}}_{2,25 \text{ years}}\right)$$

$$\left(1 + \frac{i^{(4)}}{4}\right)^5 = \left(1 + \frac{0,16}{4}\right)^5$$
$$= (1,04)^5 \rightarrow \left(\text{nominal interest convertible quarterly for 15 months (= 5 quarters)}\right)$$

i^* = Equivalent effective rate of interest for seven years

$$(1+i)^7 = (1,1) \cdot (0,96)^{-3} \cdot (0,775)^{-1} \cdot (1,04)^5$$

$$(1+i)^7 = 1,991838949$$

$$1+i = 1,10025157$$

$$i = 0,10025157$$

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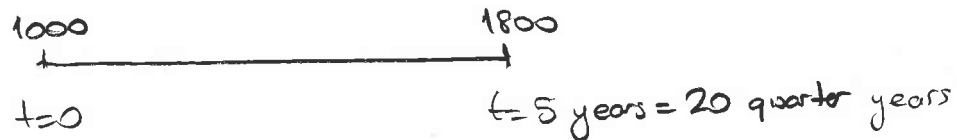
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4) convertible quarterly $\Rightarrow \left(1 + \frac{i^{(4)}}{4}\right)^t$



$$1800 = 1000 \cdot \left(1 + \frac{i(4)}{4}\right)^{20}$$

$$1 + \frac{i(4)}{4} = (1.8)^{\frac{1}{20}} = 1.029825462$$

$$i'(u) = 0,119301848$$

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$$5) \text{ convertible monthly } \rightarrow \left(1 + \frac{i^{(12)}}{12}\right)^t = \left(1 + \frac{0.12}{12}\right)^t = (1.01)^t.$$

$$10 \text{ years} = 120 \text{ months}$$

$$100000 \cdot (1.01)^{120} = 330038.6899$$

$$\sum_{n=1}^{120} 1000 \cdot (1.01)^n = 1000 \cdot \sum_{n=1}^{120} (1.01)^n = 1000 \cdot \frac{(1.01)^{120} - 1}{(1.01)^{120} \cdot (1.01 - 1)}$$

$$= \frac{230038.6899}{0.01}$$

$$= 69700,92203$$

$$330038.6899 - 69700,92203 \approx 30,299$$